Universitat de València
Escola Tècnica Superior d'Enginyeria
Departament d'Informàtica

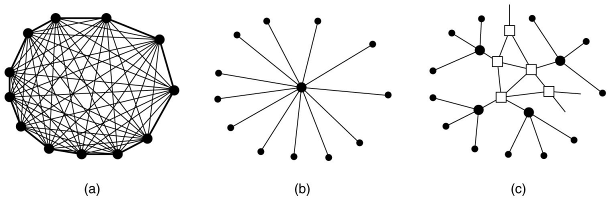
Sistema telefónico. Interfaces de la capa física

Juan Manuel Orduña Huertas

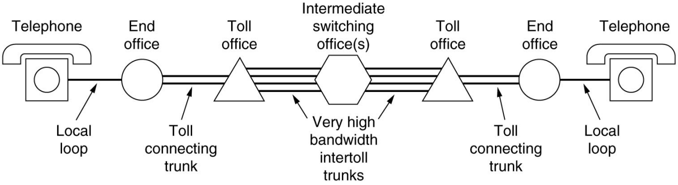
Redes de Transmisión de Datos - Curso 2011/2012

Contenido

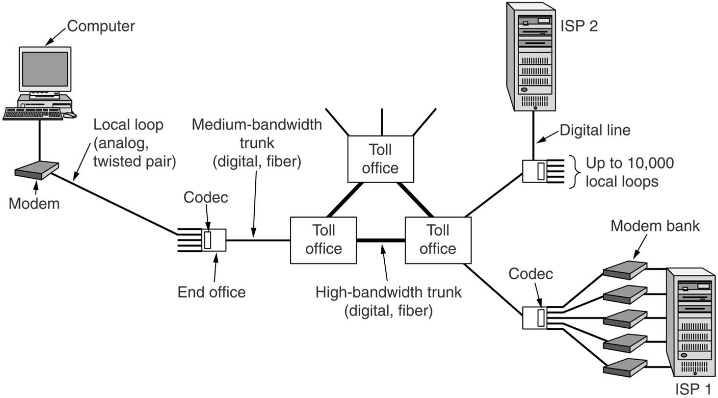
Estructura del Sistema Telefónico



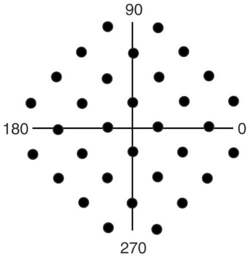
Esquema de llamada de media distancia



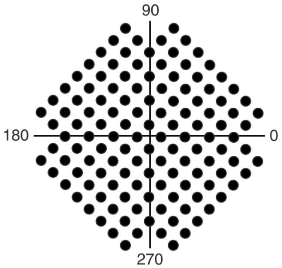
Bucles de abonado para transmisión de datos



Módems: Diagramas constelación de V.32

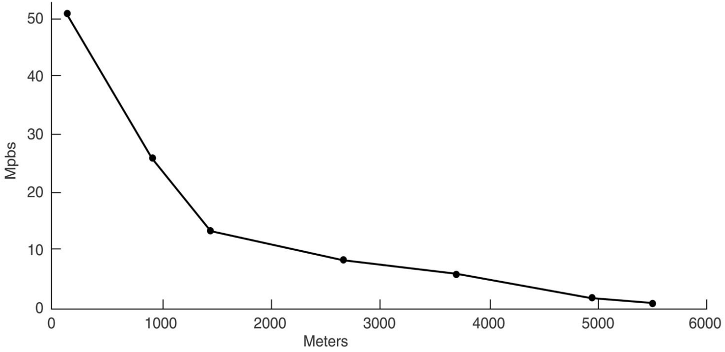


(b)

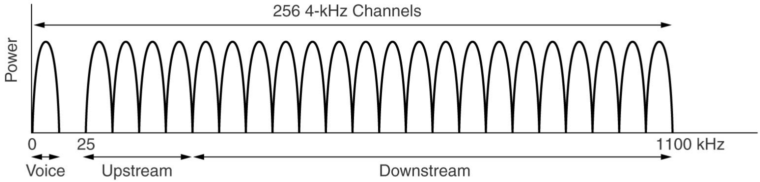


(c)

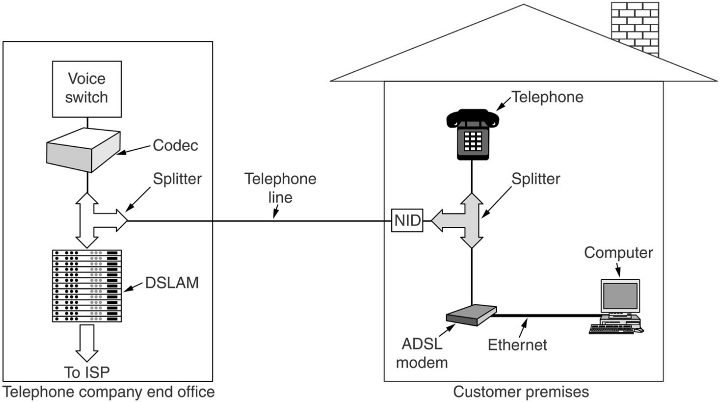
Ancho de banda potencial de bucle de abonado con cable UTP categoría 3



Asignación del ancho de banda ADSL



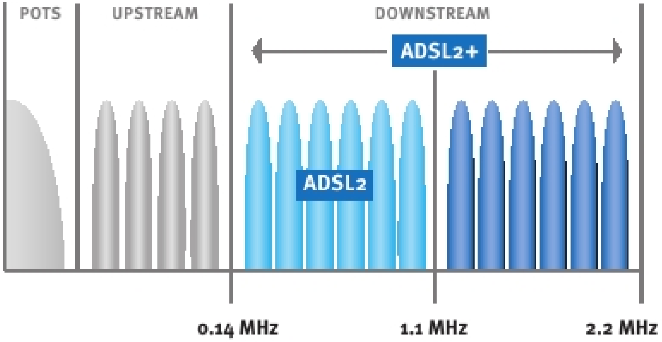
Configuración típica de sistema ADSL



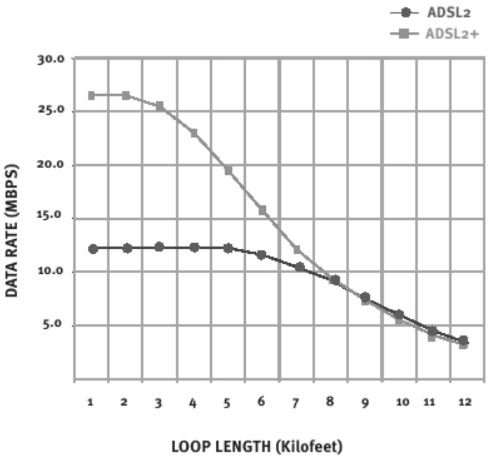
Prestaciones de los estándares ADSL

	ADSL	ADSL2	ADSL2+
Frecuencia	0,5 MHz	1,1 MHz	2,2 MHz
Vel. Trans. ascendente	1 Mbps	1 Mbps	1,2 Mbps
Vel. Max. descendente	8 Mbps	12 Mbps	24 Mbps
Distancia	2 Km	2,5 Km	2,5 km
Tiempo Sincronización	10-30 s	3 s	3 s
Corrección de Errores	No	Sí	Sí

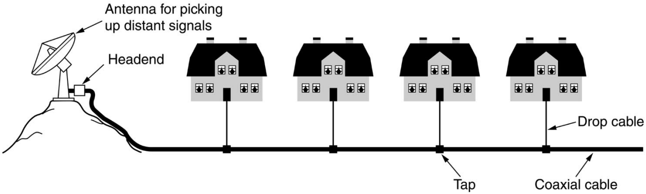
Espectro utilizado por ADSL2+



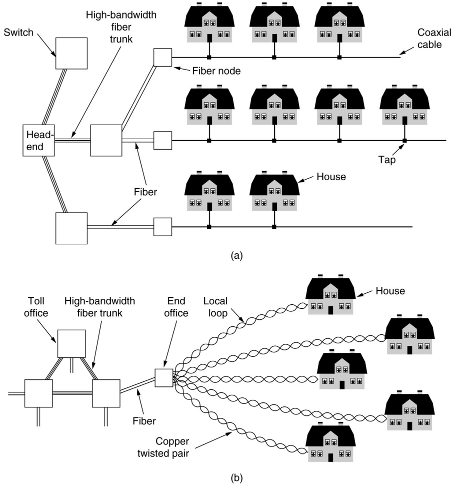
Velocidad de transmisión para diferentes distancias en ADSL2+



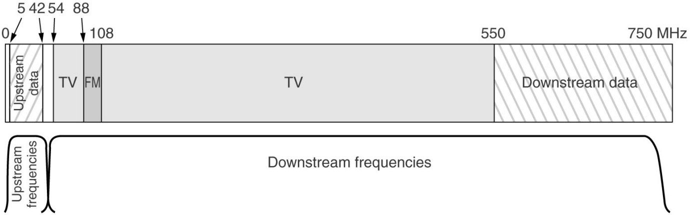
Antiguo sistema de televisión por cable



Red HFC (Hybrid Fiber-Coax)



Asignación típica de frecuencias para acceso a Internet en CATV

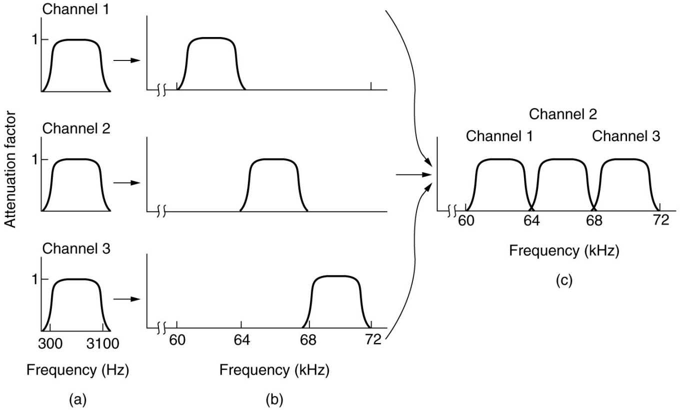


Módems de cable

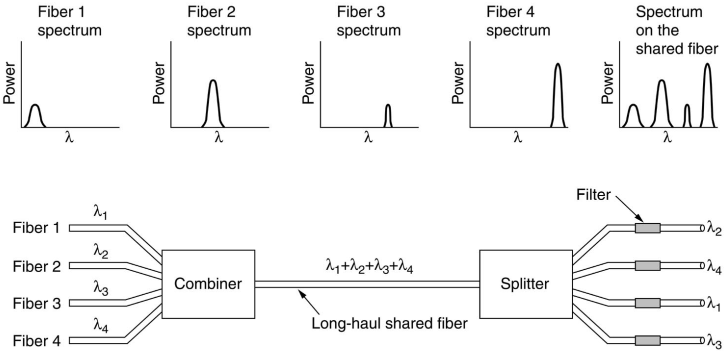
Características: - Modulación analógica, cada canal ascencentede 6 u 8 MHz se modula en QAM-64 o QAM-256.

- DOCSIS (Data-Over-Cable Service Interface Specification).
- DAVIC (Digital Audio-Visual Council).
- IEEE 802.14.

FDM: Frecuency División Multiplexing



WDM: Wavelength Division Multiplexing

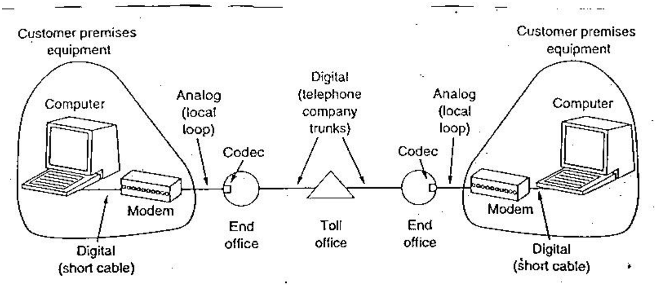


TDM: Time Division Multiplexing

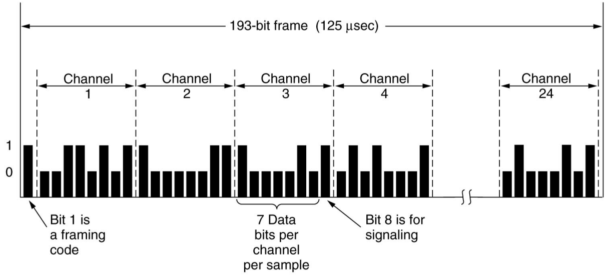
Canal digital básico:
- Modulación PCM con 8 bits - Canal analógico de 4 kHz —>
Fm mínima = 8000 muestras/seg

$$8000\ muestras/s \ * \ 8\ bits/muestra \ = \ 64 kbps$$

Transmisión de datos por RTC

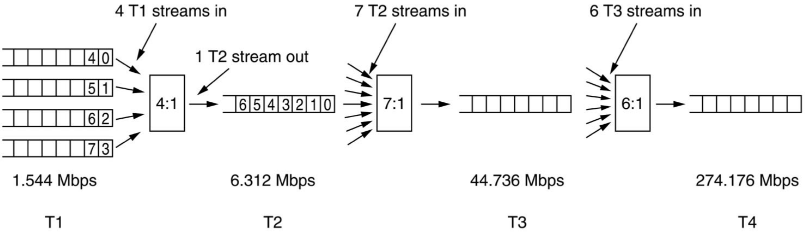


Portadora T1 - 24 canales PCM básicos

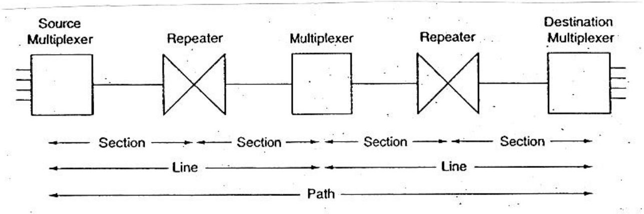


- T1: Cada 125 μs (1/8000 Hz) saca 192 bits (24x8 bits) +1 de sincronización = 1,544 Mbps
- E1: 32 canales PCM, es decir, 2,048 Mbps. 30 canales para información y 2 control

Niveles de multiplexación en troncales digitales



Configuración SONET/SDH



Topología SONET

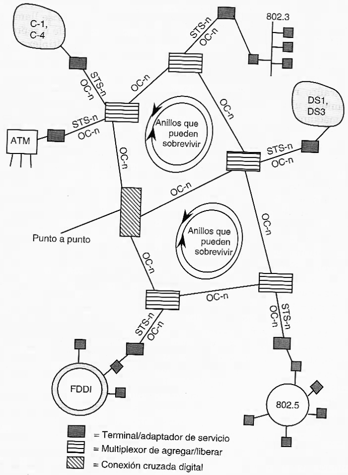
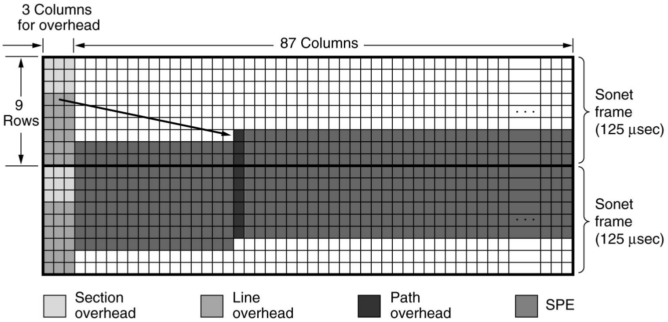


Figura 8–1 Topología SONET/SDH.

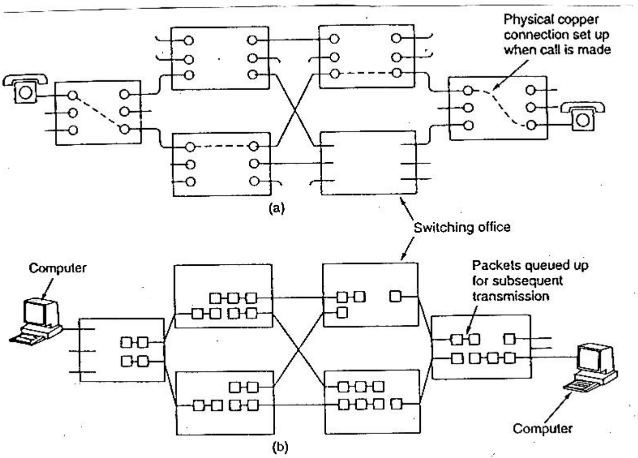
Tramas SONET/SDH consecutivas



Jerarquía de multiplexación SONET

SONET		SDH	Data rate (Mbps)		
Electrical	**Optical**	**Optical**	**Gross**	**SPE**	**User**
STS-1	OC-1		51.84	50.112	49.536
STS-3	OC-3	STM-1	155.52	150.336	148.608
STS-9	OC-9	STM-3	466.56	451.008	445.824
STS-12	OC-12	STM-4	622.08	601.344	594.432
STS-18	OC-18	STM-6	933.12	902.016	891.648
STS-24	OC-24	STM-8	1244.16	1202.688	1188.864
STS-36	OC-36	STM-12	1866.24	1804.032	1783.296
STS-48	OC-48	STM-16	2488.32	2405.376	2377.728
STS-192	OC-192	STM-64	9953.28	9621.504	9510.912

Transmisión de la información por las centrales de conmutación



Estructura general

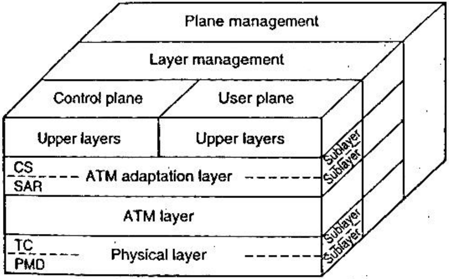
Broadband-ISBN: Diseñada para soportar

- Vídeo (televisión) a domicilio.
- Videoconferencia.
- Correo electrónico multimedia con movimiento.
- Música con calidad de CD.
- Interconexión de redes de área local.

Velocidades básicas:

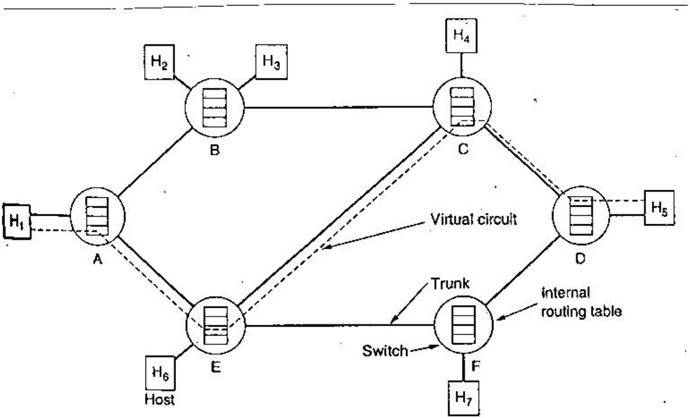
- 155,52 Mbps.
- 622 Mbps (4 canales de 155,52 Mbps multiplexados).

Arquitectura de protocolos ATM



CS: Convergence sublayer
SAR: Segmentation and reassembly sublayer
TC: Transmission convergence sublayer
PMD: Physical medium dependent sublayer

Circuito virtual

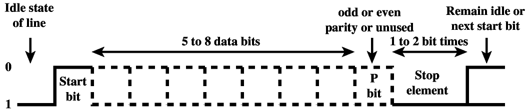


Descripción

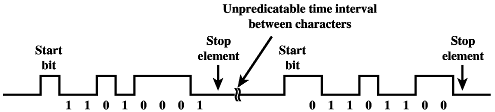
Clases de especificaciones:

1. **Eléctricas:**
2. **Mecánicas**
3. **Funcionales**
4. **Procedurales**

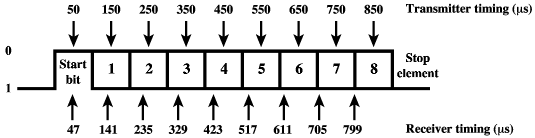
Sincronización a nivel de bit



(a) Character format



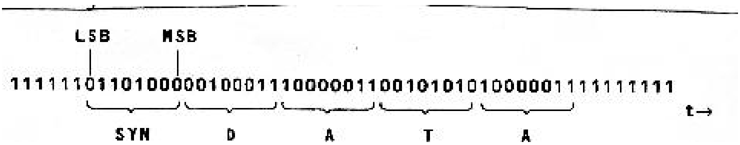
(b) 8-bit asynchronous character stream



(c) Effect of timing error

Transmisión asíncrona

Transmisión síncrona



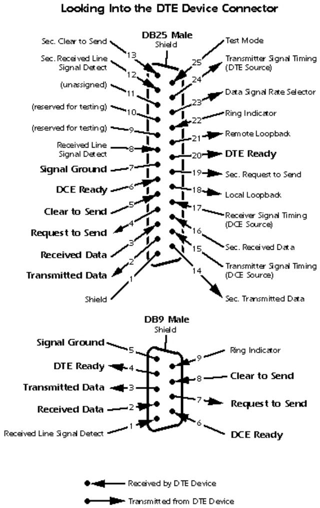
Características

1. Especifica un cable de longitud máxima de 15 m. para una velocidad máxima de transmisión de 19.200 bps.
2. Permite transmisión síncrona o asíncrona.
3. Opera sobre líneas telefónicas dedicadas o sobre la red telefónica conmutada.
4. Permite un modo de operación half-duplex o full-duplex.

Nomenclatura

RS-232-C			EIA-232-D		
Pin	*CCITT Number*	*EIA Name*	*Pin*	*CCITT Number*	*EIA Name*
1	101	AA	1		
7	102	AB	7	102	AB
2	103	BA	2	103	BA
3	104	BB	3	104	BB
4	105	CA	4	105	CA
5	106	CB	5	106	CB
6	107	CC	6	107	CC
20	108.2	CD	20	108.2	CD
22	125	CE	22	125	CE
8	109	CF	8	109	CF
21	110	CG	21	140/110	RL/CG[1]
23	111/112	CH/CI	23	111/112	CH/CI[2]
24	113	DA	24	113	DA
15	114	DB	15	114	DB
17	115	DD	17	115	DD
14	118	SBA	14	118	SBA
16	119	SBB	16	119	SBB
19	120	SCA	19	120	SCA
13	121	SCB	13	121	SCB
12	122	SCF	12	122/112	SCF/CI[3]
			9	—	—
			10	—	—
			11	—	—
			18	141	LL
			25	142	TM

Patillaje


Looking Into the DTE Device Connector
DB25 Male
Shield
Sec. Clear to Send
Sec. Received Line Signal Detect
(unassigned)
(reserved for testing)
(reserved for testing)
Received Line Signal Detect
Signal Ground
DCE Ready
Clear to Send
Request to Send
Received Data
Transmitted Data
Shield
Test Mode
Transmitter Signal Timing (DTE Source)
Data Signal Rate Selector
Ring Indicator
Remote Loopback
DTE Ready
Sec. Request to Send
Local Loopback
Receiver Signal Timing (DCE Source)
Sec. Received Data
Transmitter Signal Timing (DCE Source)
Sec. Transmitted Data
DB9 Male
Shield
Signal Ground
DTE Ready
Transmitted Data
Received Data
Received Line Signal Detect
Ring Indicator
Clear to Send
Request to Send
DCE Ready
Received by DTE Device
Transmitted from DTE Device

Función de cada patilla

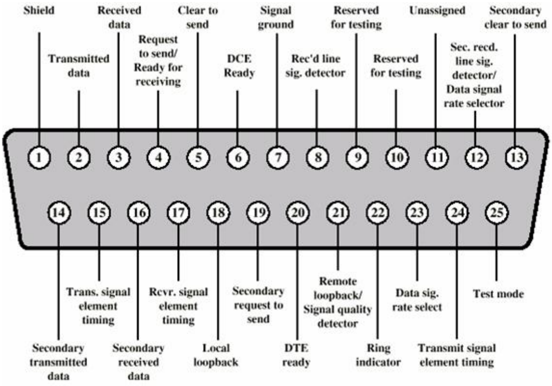
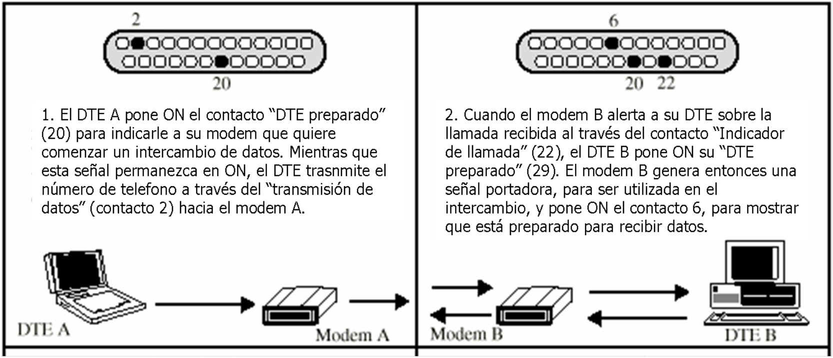
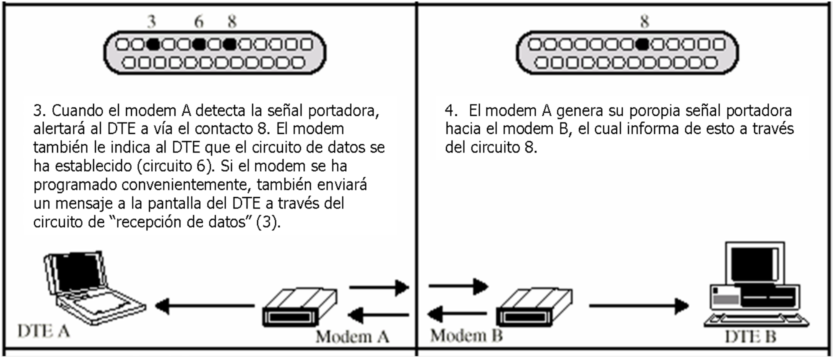


Figure 6.5 Pin Assignments for V.24/EIA-232 (DTE Connector Face)

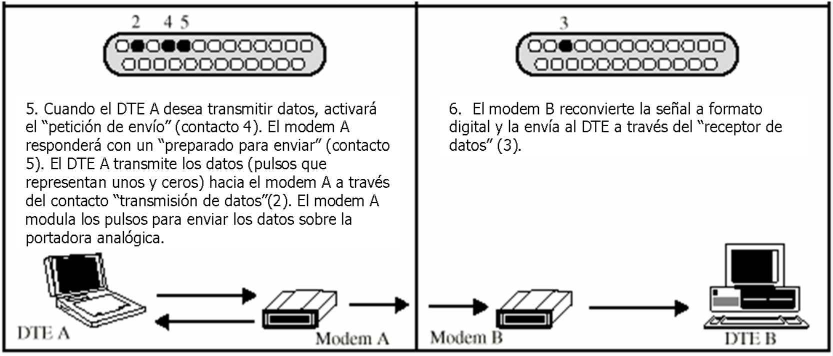
Características Procedurales I



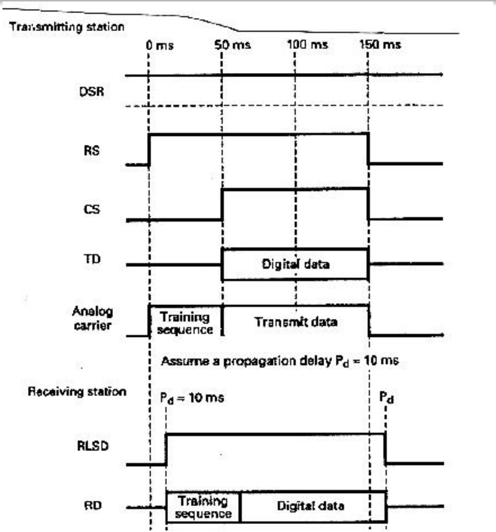
Características Procedurales II



Características Procedurales III



Modo de operación básico



Otra norma: V.35

Nombre	Patilla	Descripción	Tipo
FG	A	Frame Ground	–
SG	B	Signal Ground	–
SDA	P	Send Data A	Diferencial
SDB	S	Send Data B	Diferencial
RDA	R	Receive Data A	Diferencial
RDB	T	Receive Data B	Diferencial
RTS	C	Request To Send	No balanceada (V.24)
CTS	D	Clear To Send	No balanceada
DSR	E	Data Set Ready	No balanceada
DTR	H	Data Terminal Ready	No Balanceada
RLSD	F	Receive Line Signal Detect	No Balanceada
TCEA	U	Transmit Clock Ext A	Diferencial
TCEB	W	Transmit Clock Ext B	Diferencial
TCA	Y	Transmit Clock A	Diferencial
TCB	AA	Transmit Clock B	Diferencial
RCA	VReceive Clock A	Diferencial	
RCB	X	Receive Clock B	Diferencial
LL	J	Local Loopback	No Balanceada
RLB	BB	Remote Loopback	No Balanceada
TM	K	TestMode	No Balanceada
	L	TestPattern	No balanceada

Bibliografía I

- Andrew S. Tanenbaum, *"Redes de Computadores", Ed. Prentice Hall, 4a. ed., 2003*
- William Stallings, *"Data and Computr Communications", Ed. Prentice Hall, 8ª ed., 2007*